MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3450 CCD & POX PROCESS WATER STORAGE FOUNDATION DESIGN OF 3450-TNK-001

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
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QA	:	Quality Assurance
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Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to Prepare for design of **CCD & POX PROCESS WATER STORAGE (3450-TNK-001)** Tank Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in excel and STAAD PRO. For member strength calculation force values are directly taken from load data provided and output of STAAD analysis.

4. DESIGN REQUIREMENTS

4.1 Design Loads

All Design Loads are indicated in Document No. MD-216-3450-EG-ST-GAD- 2245 attached in Attachment 2. Also indicated in Civil, Structural and Architectural Design Criteria (Document No MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

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5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations for above ground Tanks

Foundation shall be Raft Foundation with a Concrete Ringwall.

5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Wall	50
Exposed to weather or contact with ground	Raft	50

5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 300 kN/m^2 at 1.25m depth below EGL. (refer Table 5 - BH-9 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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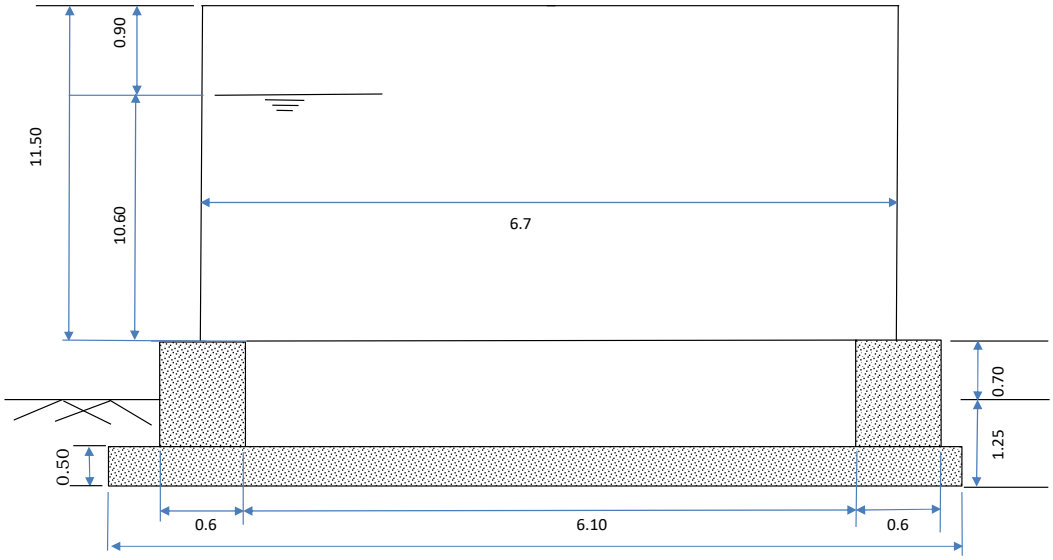
Foundation for 3450-TNK-001

Tank Load Data

DEAD WEIGHT OF TANK	322	kN
ROOF PIPE LIQUID LOAD	53	kN
SHELL PIPE LIQUID LOAD	53	kNm
TANK LIQUID LOAD (OPERATING)	3656	kN
TANK LIQUID LOAD (TEST)	3977	kN
LIVE LOAD	149	kN
WIND LOAD - HORIZONTAL	42	kN
WIND LOAD - VERTICAL	-35	kN
SEISMIC LOAD	61	kN

	Loading Type		
Weight, V	Empty (W)	322	kN
	Operating Condition (Wo)	4084	kN
	Test Condition (Wt)	4405	kN

Summary of Tank Specification



Diameter of Tank, D	=	6.7 m	
Height of Tank, H	=	11.5 m	
Height of Tank Fluid, H_L	=	10.6 m	
Inner Diameter of Concrete Ring Wall, ID	=	6.1 m	
Outer Diameter of Concrete Ring Wall, OD	=	7.3 m	
Height of Concrete Ring Wall, h	=	1.95 m	
Thickness of Concrete Ring Wall, t	=	0.6 m	
Thickness of Concrete Raft, T	=	0.5 m	
Top of Foundation	=	0.7 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L	=	10 kN/m^3	
Unit Weight of Soil, γ_s	=	20 kN/m^3	
Unit Weight of Concrete, γ_c	=	24 kN/m^3	

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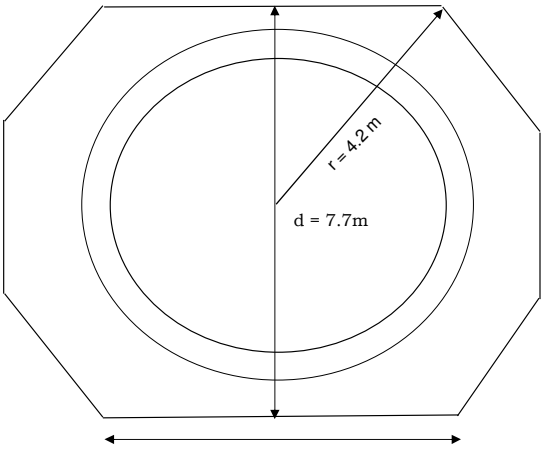
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Wind Load & Seismic Load

Total Base Shear due to wind load = 42 kN

Total Base Shear due to Seismic load = 61 kN

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	42.00	368.00
	Operating	42.00	368.00
Seismic Loads	Operating	61.00	291.00



Foundation Weight

a) Ring Wall Weight = 439.50 kN

b) Inside Soil Weight = 847.52 kN

c) Outside Soil Weight = 59.62 kN

d) Raft Weight = 549.94 kN

Area of Raft

Area of Raft (Hexagonal), $A = 2.598 * r^2$ = 45.83 m^2

Property of Hexagon

Radius of outer circle, r = 4.2 m

Dist. of 2 opposite side, d = 7.7 m

Sectional Modulus of Raft

Sectional modulus (Hexagonal), $Ze = 0.0601*d^4 / r$ = 50.30 m^3

Basic Load Cases

1. Tank Dead Load (D)
2. Live Load (L)
3. Operating liquid Load (F1)
4. Test liquid Load (F2)
5. Wind Load (W)
6. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4.1)

Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Operating	LC101	D + F1	4084.00	0.00	0.00
Empty	LC102	0.6 D + W	193.20	42.00	368.00
Operating	LC103	D + F1 + W	4049.00	42.00	368.00
	LC104	D + F1+ 0.7 E	4084.00	42.70	203.70
	LC105	0.6D + 0.7 E	2450.40	42.70	203.70
Test	LC106	D + F2	4405.00	0.00	0.00
	LC107	D + F2 + L	4554.00	0.00	0.00
	LC108	D + F2 + W	4370.00	42.00	368.00
	LC109	D + F2 + 0.7 E	4405.00	42.70	203.70
Operating	LC110	D + F1 + L	4233.00	0.00	0.00

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Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 300 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z	Qmax= V/A - M/Z	Qall/ Qmax	Soil Bearing Check
	V (kN)	M(kNm)	kN/m^2	kN/m^2		
LC101	5980.58	0.00	130.50	130.50	2.30	OK
LC102	2089.78	449.90	54.54	36.66	5.50	OK
LC103	5945.58	449.90	138.68	120.79	2.16	OK
LC104	5980.58	286.97	136.20	124.79	2.20	OK
LC105	4346.98	286.97	100.56	89.15	2.98	OK
LC106	6301.58	0.00	137.50	137.50	2.18	OK
LC107	6450.58	0.00	140.75	140.75	2.13	OK
LC108	6266.58	449.90	145.68	127.80	2.06	OK
LC109	6301.58	286.97	143.21	131.80	2.09	OK
LC110	6129.58	0.00	133.75	133.75	2.24	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of V (kN)	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
LC101	5980.58	0.3	1794.17	0.00	-	-
LC102	2089.78	0.3	626.93	42.00	14.9	OK
LC103	5945.58	0.3	1783.67	42.00	42.5	OK
LC104	5980.58	0.3	1794.17	42.70	42.0	OK
LC105	4346.98	0.3	1304.09	42.70	30.5	OK
LC106	6301.58	0.3	1890.47	0.00	-	-
LC107	6450.58	0.3	1935.17	0.00	-	-
LC108	6266.58	0.3	1879.97	42.00	44.8	OK
LC109	6301.58	0.3	1890.47	42.70	44.3	OK
LC110	6129.58	0.3	1838.87	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of V (kN)	Edge Dis.(d/2)	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
LC101	5980.58	3.85	23025.23	0.00	-	-
LC102	2089.78	3.85	8045.65	449.90	17.9	OK
LC103	5945.58	3.85	22890.48	449.90	50.9	OK
LC104	5980.58	3.85	23025.23	286.97	80.2	OK
LC105	4346.98	3.85	16735.87	286.97	58.3	OK
LC106	6301.58	3.85	24261.08	0.00	-	-
LC107	6450.58	3.85	24834.73	0.00	-	-
LC108	6266.58	3.85	24126.33	449.90	53.6	OK
LC109	6301.58	3.85	24261.08	286.97	84.5	OK
LC110	6129.58	3.85	23598.88	0.00	-	-

Ultimate Strength Design

Load Combination (As per SBC-304 , Table 5.3.1)

Load Conditions	Load Case	Load Combination	Load at bottom of tank Inner Area			
			V (kN)	M (kNm)	A (m^2)	Z (m^3)
Operating	LC201	1.4 (D + F1)	5717.60	0.00	35.26	29.53
	LC202	1.4 (D + F1) + 1.7 L	5970.90	0.00	35.26	29.53
Empty	LC203	0.9 D + 1.6 W	233.80	588.80	35.26	29.53
Operating	LC204	1.2(D + F1) + 1.6W + L	4993.80	588.80	35.26	29.53
	LC205	1.2 (D + F1) + E + L	5049.80	291.00	35.26	29.53
	LC206	0.9 (D + F1) + E	3675.60	291.00	35.26	29.53
Test	LC207	1.4 (D + F2)	6167.00	0.00	35.26	29.53
	LC208	1.4 (D + F2) + 1.7 L	6420.30	0.00	35.26	29.53
	LC209	1.2(D + F2) + 1.6W + L	5379.00	588.80	35.26	29.53
	LC210	1.2 (D + F2) + E + L	5435.00	291.00	35.26	29.53

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Staad input calculation.

Total Number of Noads in Wall Top = 24

S/W of tank (Empty Condition) on each Nodes= 13.42 kN

Pressure due to Sand fill on inside Raft= 29.0 kN/m^2

Pressure due to soil fill on outside Raft= 15.0 kN/m^3

Wind Base Shear of tank on each Nodes= 1.750 kN

Seismic Base Shear of tank on each Nodes= 2.542 kN

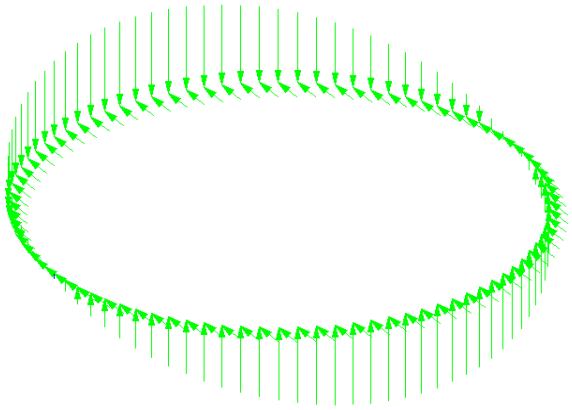
Sl No.	Node No.	Dist. R(m)	R^2	Seismic V(kN)	Wind V(kN)
1	3452	-3.4	11.22	-7.2	-9.2
2	3453	-3.2	10.47	-7.0	-8.8
3	3454	-2.9	8.42	-6.3	-7.9
4	3455	-2.4	5.61	-5.1	-6.5
5	3456	-1.7	2.81	-3.6	-4.6
6	3457	-0.9	0.75	-1.9	-2.4
7	3458	0.0	0.00	0.0	0.0
8	3459	0.9	0.75	1.9	2.4
9	3460	1.7	2.81	3.6	4.6
10	3461	2.4	5.61	5.1	6.5
11	3462	2.9	8.42	6.3	7.9
12	3463	3.2	10.47	7.0	8.8
13	3464	3.4	11.22	7.2	9.2
14	3465	3.2	10.47	7.0	8.8
15	3466	2.9	8.42	6.3	7.9
16	3467	2.4	5.61	5.1	6.5
17	3468	1.7	2.81	3.6	4.6
18	3469	0.9	0.75	1.9	2.4
19	3470	0.0	0.00	0.0	0.0
20	3471	-0.9	0.75	-1.9	-2.4
21	3472	-1.7	2.81	-3.6	-4.6
22	3473	-2.4	5.61	-5.1	-6.5
23	3474	-2.9	8.42	-6.3	-7.9
24	3475	-3.2	10.47	-7.0	-8.8
		ΣR^2=	**134.67**		

Staad input calculation For Seismic and Wind load distribution.

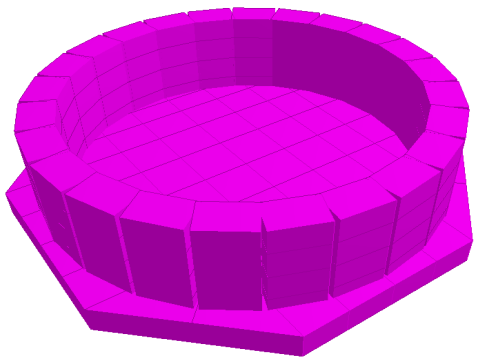
$V_{wind} = R * W_{total} / \Sigma R^2$ $H_{wind} = W_{total} / 24$

$V_{Seismic} = R * E_{total} / \Sigma R^2$ $H_{seismic} = E_{total} / 24$

H_{wind}=	1.750	kN
$H_{seismic}$=	2.542	kN



Staad Model in 3D view



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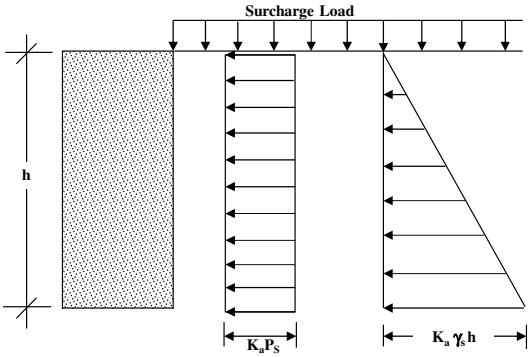
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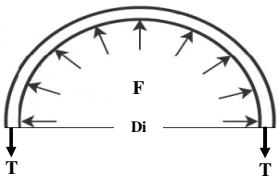
Wall Design

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m^2
LC201	5717.60	0.00	162.17
LC202	5970.90	0.00	169.36
LC203	233.80	588.80	26.57
LC204	4993.80	588.80	161.58
LC205	5049.80	291.00	153.09
LC206	3675.60	291.00	114.11
LC207	6167.00	0.00	174.92
LC208	6420.30	0.00	182.10
LC209	5379.00	588.80	172.51
LC210	5435.00	291.00	164.01

Maximum Surcharge Load, Ps = 182.10 kN/m^2





Where, K_a = coefficient of earth pressure at rest
= 0.5

Pressure due to Surcharge,P1 = $K_a * P_s$ = 91.05 KN/m^2 LC208 - 1.4 (D + F2) + 1.7 L

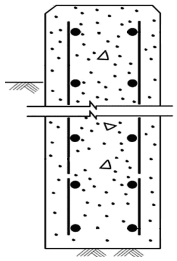
Pressure due to Soil,P2 = $1.4 * \gamma_S * (h-T) * K_a$ = 20.30 KN/m^2

Maximum Lateral Load

$F = P1 * (h-T) + 1/2 * P2 * (h-T)$ = 146.7418 KN/m

Hoop Tension Load

$T = 1/2 * F * ID$ = 447.5625 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = $T/(\phi \times Fy)$ = 2175 mm^2

Minimum As = 0.0025 x t x h = 2175 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 16 Nos. D16= 3216.991 mm^2 OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 900 mm^2 (As per API 650 , Cl. No. B.4.2.3)

STAAD output:- (Refer Attachment 1)

Wall Thickness = 600 mm

Characteristic strength of concrete, f'_c = 35 MPa

Yield stress of steel reinforcement, f_y = 420 MPa

Clear cover for main reinforcement bars = 50mm

Max BM (+) (M_y)= 6.845 kN m/m (VERTICAL MOMENT IN OUTSIDE FACE)

Max BM (-) (M_y)= 49.007 kN m/m (VERTICAL MOMENT IN INSIDE FACE)

Max BM (+) (M_x)= 1.247 kN m/m (HORIZONTAL MOMENT IN OUTSIDE FACE)

Max BM (-) (M_x)= 8.628 kN m/m (HORIZONTAL MOMENT IN INSIDE FACE)

Max Shear Qx = 0.01 N/mm^2

Max Shear Qy = 0.15 N/mm^2

Effective depth of wall in Vertical direction = 542 mm

Effective depth of wall in Horizontal direction = 526 mm

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Moment	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
My(+)	6.85	0.02	0.03	0.000	40.11	975.60	D16	206.09	300	200	1005.31
My(-)	49.01	0.17	0.22	0.001	288.13	975.60	D16	206.09	300	200	1005.31
Mx(+)	1.25	0.00	0.01	0.000	7.53	975.60	D16	206.09	300	200	1005.31
Mx(-)	8.63	0.03	0.04	0.000	52.11	975.60	D16	206.09	300	200	1005.31

Check against one-way shear force

Allowable shear stress, (ϕ Vc)/bd = 0.74 N/mm^2

Shear stress at a distance 'd' from raft top , τ_v= 0.15 N/mm^2 **OK**

SUMMARY OF REINFORCEMENT IN WALL: **600 mm**

VERTICAL REIN. IN INSIDE FACE	**D16**	**200**	**c/c**	
VERTICAL REIN. IN OUTSIDE FACE	**D16**	**200**	**c/c**	
HORIZONTAL REIN. IN INSIDE FACE	**D16**	**200**	**c/c**	**(8Nos.)**
HORIZONTAL REIN. IN OUTSIDE FACE)	**D16**	**200**	**c/c**	**(8Nos.)**

Raft Design

STAAD output:- (Refer Attachment 1)

Raft Thickness = 500 mm

Max BM (+) (M_y)=	24.47	kN m/m	(HORIZONTAL MOMENT IN BOTTOM FACE)
Max BM (-) (M_y)=	25.394	kN m/m	(HORIZONTAL MOMENT IN TOP FACE)
Max BM (+) (M_x)=	21.62	kN m/m	(LONGITUDINAL MOMENT IN BOTTOM FACE)
Max BM (-) (M_x)=	17.126	kN m/m	(LONGITUDINAL MOMENT IN TOP FACE)

Max Shear Qx =	0.06	N/mm^2
Max Shear Qy =	0.13	N/mm^2

Effective depth of Raft in Vertical direction = 417 mm

Effective depth of Raft in Horizontal direction = 401 mm

Moment	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
My(+)	24.47	0.14	0.19	0.000	186.85	750.60	D16	267.87	300	200	1005.31
My(-)	25.39	0.15	0.19	0.000	193.96	750.60	D16	267.87	300	200	1005.31
Mx(+)	21.62	0.13	0.18	0.000	171.69	721.80	D16	278.56	300	200	1005.31
Mx(-)	17.13	0.11	0.14	0.000	135.91	721.80	D16	278.56	300	200	1005.31

Check against one-way shear force

Allowable shear stress, (ϕ Vc)/bd = 0.74 N/mm^2

Shear stress at a distance 'd' from raft top , τ_v= 0.13 N/mm^2 **OK**

SUMMARY OF REINFORCEMENT IN RAFT: **500 mm**

(HORIZONTAL REIN. IN TOP FACE)	**D16**	**200**	**c/c**
(HORIZONTAL REIN. IN BOTTOM FACE)	**D16**	**200**	**c/c**
(LONGITUDINAL REIN. IN TOP FACE)	**D16**	**200**	**c/c**
(LONGITUDINAL REIN. IN BOTTOM FACE)	**D16**	**200**	**c/c**

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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ATTACHMENT 1 PLATE CENTRE STRESS SUMMARY

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STAAD.Pro Report

To: From:

Copy to: Date: 10/06/2020 13:30:00 Ref: ca/ Document1

Plate Center Stress Summary

	Plate	L/C	Shear Qx (N/mm²)	Shear Qy (N/mm²)	Membrane Sx (N/mm²)	Membrane Sy (N/mm²)	Membrane Sxy (N/mm²)	Bending Mx (kNm/m)	Bending My (kNm/m)	Bending Mxy (kNm/m)
Max Qx	7388	208:1.4 (D + F2) +	**0.007**	-0.126	-0.004	-0.103	0.008	-6.971	-41.843	1.144
Min Qx	7407	208:1.4 (D + F2) +	**-0.007**	-0.126	-0.004	-0.103	-0.008	-6.976	-41.900	-1.124
Max Qy	7410	203:0.9 D + 1.6 W	-0.001	**0.012**	0.044	-0.078	-0.001	0.812	3.603	-0.065
Min Qy	7408	207:1.4 (D + F2)	-0.000	**-0.145**	-0.005	-0.112	0.001	-8.627	-49.007	-0.096
Max Sx	7465	208:1.4 (D + F2) +	-0.000	0.001	**0.440**	-0.064	-0.000	-0.550	-1.901	-0.029
Min Sx	7409	207:1.4 (D + F2)	0.007	-0.133	**-0.009**	-0.080	0.010	-7.571	-45.343	0.988
Max Sy	7470	203:0.9 D + 1.6 W	0.000	0.007	0.026	**0.000**	-0.001	0.053	0.743	0.048
Min Sy	7387	209:1.2 (D + F2) +	0.000	-0.121	-0.004	**-0.141**	0.003	-7.178	-40.595	-0.059
Max Sxy	7391	209:1.2 (D + F2) +	0.006	-0.110	-0.005	-0.085	**0.022**	-6.103	-36.674	0.603
Min Sxy	7404	209:1.2 (D + F2) +	-0.006	-0.110	-0.005	-0.085	**-0.022**	-6.132	-36.809	-0.599
Max Mx	7409	203:0.9 D + 1.6 W	0.002	0.009	0.009	-0.088	0.001	**1.247**	6.844	0.349
Min Mx	7387	207:1.4 (D + F2)	0.000	-0.145	-0.005	-0.112	-0.001	**-8.628**	-48.984	0.100
Max My	7386	203:0.9 D + 1.6 W	-0.002	0.009	0.009	-0.088	-0.001	1.247	**6.845**	-0.346
Min My	7408	207:1.4 (D + F2)	-0.000	-0.145	-0.005	-0.112	0.001	-8.627	**-49.007**	-0.096
Max Mxy	7406	209:1.2 (D + F2) +	0.006	-0.108	-0.006	-0.102	-0.002	-6.064	-36.579	**1.145**
Min Mxy	7407	208:1.4 (D + F2) +	-0.007	-0.126	-0.004	-0.103	-0.008	-6.976	-41.900	**-1.124**

STAAD.Pro Report

To: From:

Copy to: Date: 10/06/2020 13:35:00 Ref: ca/ Document1

Plate Center Stress Summary

	Plate	L/C	Shear Qx (N/mm²)	Shear Qy (N/mm²)	Membrane Sx (N/mm²)	Membrane Sy (N/mm²)	Membrane Sxy (N/mm²)	Bending Mx (kNm/m)	Bending My (kNm/m)	Bending Mxy (kNm/m)
Max Qx	7741	208:1.4 (D + F2) +	**0.110**	0.062	0.000	0.000	0.000	21.574	6.742	8.218
Min Qx	7726	208:1.4 (D + F2) +	**-0.110**	0.062	0.000	0.000	0.000	21.496	6.732	-8.216
Max Qy	7723	208:1.4 (D + F2) +	-0.061	**0.110**	0.000	0.000	0.000	6.710	21.606	-8.119
Min Qy	7732	208:1.4 (D + F2) +	-0.016	**-0.125**	0.000	0.000	0.000	2.478	24.466	0.516
Max Sx	7743	201:1.4 (D + F1)	-0.015	0.080	**0.000**	0.000	0.000	9.191	14.546	-5.695
Min Sx	7743	201:1.4 (D + F1)	-0.015	0.080	**0.000**	0.000	0.000	9.191	14.546	-5.695
Max Sy	7743	201:1.4 (D + F1)	-0.015	0.080	0.000	**0.000**	0.000	9.191	14.546	-5.695
Min Sy	7743	201:1.4 (D + F1)	-0.015	0.080	0.000	**0.000**	0.000	9.191	14.546	-5.695
Max Sxy	7743	201:1.4 (D + F1)	-0.015	0.080	0.000	0.000	**0.000**	9.191	14.546	-5.695
Min Sxy	7743	201:1.4 (D + F1)	-0.015	0.080	0.000	0.000	**0.000**	9.191	14.546	-5.695
Max Mx	7738	208:1.4 (D + F2) +	0.109	-0.062	0.000	0.000	0.000	**21.621**	6.531	-8.103
Min Mx	7640	207:1.4 (D + F2)	-0.070	-0.005	0.000	0.000	0.000	**-17.126**	2.809	-1.941
Max My	7732	208:1.4 (D + F2) +	-0.016	-0.125	0.000	0.000	0.000	2.478	**24.466**	0.516
Min My	7653	207:1.4 (D + F2)	0.004	0.078	0.000	0.000	0.000	-1.683	**-25.394**	-0.839
Max Mxy	7659	207:1.4 (D + F2)	0.040	-0.055	0.000	0.000	0.000	-11.072	-15.613	**11.527**
Min Mxy	7662	207:1.4 (D + F2)	-0.040	-0.055	0.000	0.000	0.000	-11.089	-15.576	**-11.515**

ATTACHMENT 2 INPUT DRAWINGS

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DRAWING NUMBER: MD-216-3450-EG-ST-GAD-2245 | REV: D01

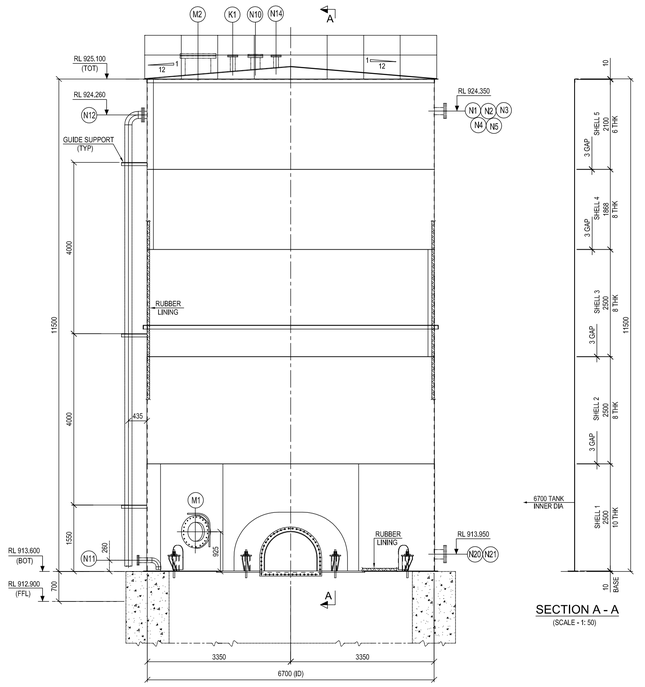
MATERIAL SPECIFICATION	
SHELL, BOTTOM, ANNULAR, ROOF	EN 10025 GR.S275 J0
COD	EN 10025 GR.S275 J0
MANWAY NECKS/ MANWAY FLANGES	EN 10025 GR.S275 J0
NOZZLE NECKS (PIPES) (600 NB & BELOW)	ASTM A53 Gr.B/ ASTM A106 Gr.B
NOZZLE NECKS (PIPES) (> 600 NB)	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
NOZZLE FLANGES/ BLIND FLANGE	SA 105
GASKETS FOR NOZZLES	FLAT RING 1.5 MM/3.00 MM THICK TANGED GRAPHITE
GASKETS FOR MANWAY/ COD	3mm THK, COMPRESSED NON-ASBESTOS SHEETS,
BOLTS/ NUTS FOR NOZZLE	SA 193 Gr. B7 / 194 Gr. 2H
BOLTS/ NUTS FOR MANWAY	SA 193 Gr. B7 / 194 Gr. 2H
STRUCTURAL BOLTS	ASTM A325M
ANCHOR CHAIR	EN 10025 GR.S275 J0
STRUCTURAL(ROLLED SECTION)+STAIRWAY / PLATFORM / ROOF SUPPORTING STRUCTURE	EN 10025 GR.S275 JR
HANDRAIL PIPE / HANDRAIL POST	ASTM A53 TYPE E
PIPE FITTINGS	SA 234 Gr. WPB
REINFORCEMENT PAD/GUSSET/ EXTERNAL SUPPORT ETC.	EN 10025 GR.S275 J0
EARTHING LUGS	EN 10025 GR.S275 J0
ANCHOR BOLTS	ASTM F1554 GR.36
CURB ANGLE	EN 10025 GR.S275 JR
NAME PLATE	STAINLESS STEEL(SS 316)

FOUNDATION LOAD DATA: **					
BASIC LOAD CASE	VERTICAL COMPRESSION	VERTICAL UPLIFT	HORIZONTAL SHEAR	OVERTURNING MOMENT (ABOUT SHELL BOTTOM JOINT)	SLAB MOMENT
	(KN)	(KN)	(KN)	(KN-M)	(KN-M)
DEAD LOAD (NOMINAL THICKNESS)	322	--	--	--	--
DEAD LOAD (CORRODED THICKNESS)	272	--	--	--	--
ROOF PIPE LIQUID LOAD	53	--	--	--	--
SHELL PIPE LIQUID LOAD	53	--	--	--	--
LIVE LOAD	149	--	--	--	--
TANK LIQUID LOAD (OPERATING)	3656	--	--	--	--
TANK LIQUID LOAD (TEST)	3977	--	--	--	--
WIND LOAD ON SHELL	--	--	42	249	--
WIND LOAD ON ROOF	--	35	--	119	--
SEISMIC LOAD	130	130	61	291	348

** REFER NOTE 15

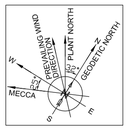
NOZZLE SCHEDULE													
NOZZLE	SERVICE	SIZE (DN)	THICKNESS (SCHEDULE)	RATING #	TYPE	ELEVATION "H"	PROJECTION "P"	RADIUS "R"	ORIENTATION	LOCATION	RF PAD DIA *	RF PAD THK. *	REMARK
N1	INLET	150	Sch XS	CL 150	RFSO	10.750	200	-	225°	SHELL	400	6	
N2	RE-CIRCULATION LINE FOR POX DEMIN WATER PUMP	80	Sch XS	CL 150	RFSO	10.750	175	-	170°	SHELL	265	6	
N3	COMBINED RE-CIRCULATION LINE FOR HP DEMIN WATER PUMP	80	Sch XS	CL 150	RFSO	10.750	175	-	180°	SHELL	265	6	
N4	RE-CIRCULATION LINE FOR HP DEMIN WATER PUMP	50	Sch XS	CL 150	RFSO	10.750	150	-	190°	SHELL	-	-	
N5	RE-CIRCULATION LINE FOR HP DEMIN WATER PUMP	50	Sch XS	CL 150	RFSO	10.750	150	-	200°	SHELL	-	-	
N10	VENT/CO2 BREATHER	150(TBC)	-	CL 150	RFSO	-	300	1000	202°	ROOF	400	6	
N11	DRAIN	100	Sch XS	CL 150	RFSO	0.260	200		90°	SHELL	305	10	
N12	OVERFLOW	150	Sch XS	CL 150	RFSO	10.660	200	-	160°	SHELL	400	6	
N13	CLEAN OUT DOOR (COD)+hxb	900x1200		AP1 650		-	-	-	130°	SHELL	2700x1372	10	
N14	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	-	400	1500	350°	ROOF	305	6	
N20	OUTLET-POX DEMIN WATER	100	Sch XS	CL 150	RFSO	0.350	200	-	160°	SHELL	305	10	
N21	OUTLET-HP DEMIN WATER	150	Sch XS	CL 150	RFSO	0.350	200	-	210°	SHELL	400	10	
K1	LEVEL TRANSMITTER	100	Sch XS	CL 150	RFSO	-	400	1500	250°	ROOF	305	6	
M1	MANHOLE C/W DAVIT	750	-	AP1 650		0.925	175	-	25°	SHELL	1545	10	
M2	MANHOLE C/W HINGED COVER	600	-			-	500	2500	293°	ROOF	1150	6	

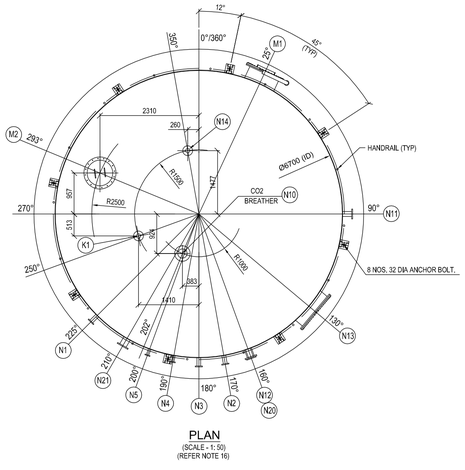
* REFER NOTE 12



TANK ELEVATION
(SCALE 1:50)

SECTION A - A
(SCALE - 1: 50)





PLAN
(SCALE - 1: 50)
(REFER NOTE 16)

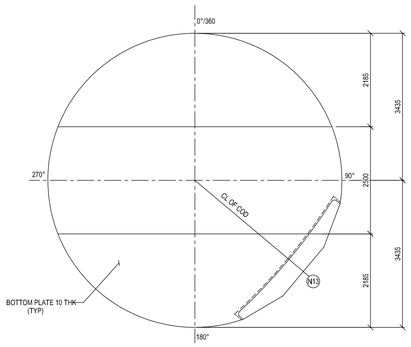
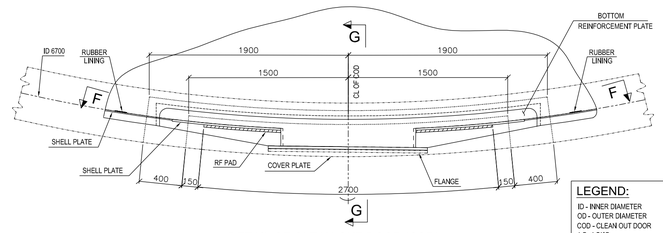
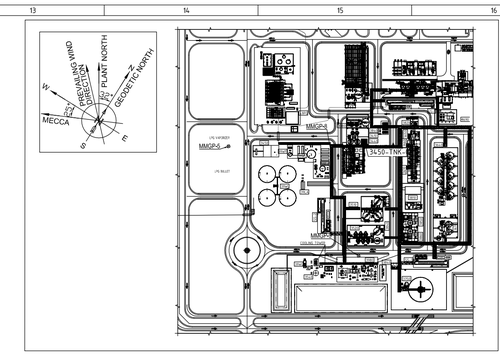


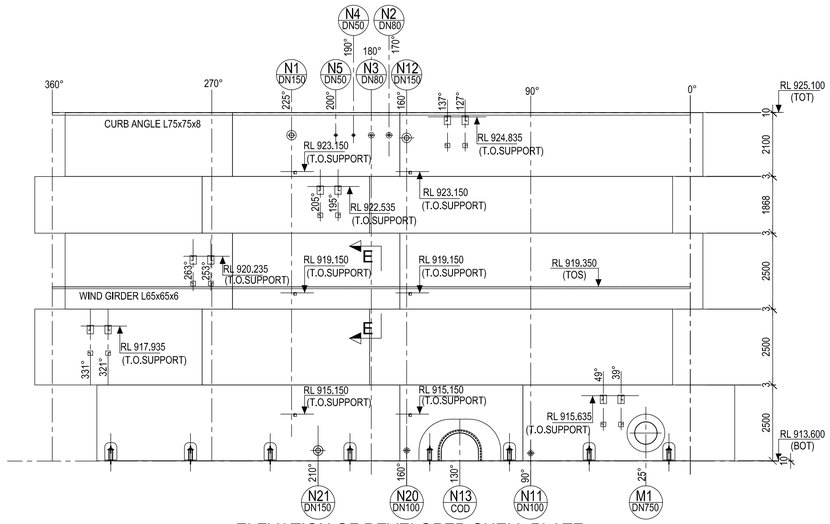
PLATE ARRANGEMENT OF BOTTOM PLATE
(SCALE - 1: 50)



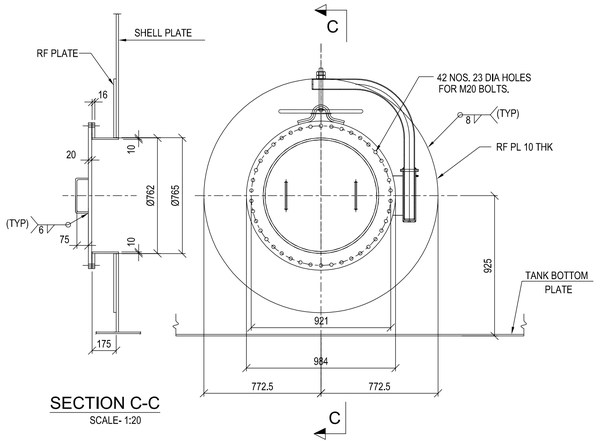
DETAIL OF CLEAN OUT DOOR (COD)
(SIZE 900x1200)
SCALE- 1:20



DRAWING NUMBER MD-216-3450-EG-ST-GAD-2246 REV D01

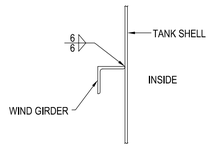


ELEVATION OF DEVELOPED SHELL PLATE
(FOR TANK NO 3450-TNK-001)
(SCALE - 1: 100)

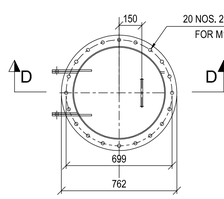


SECTION C-C
SCALE- 1:20

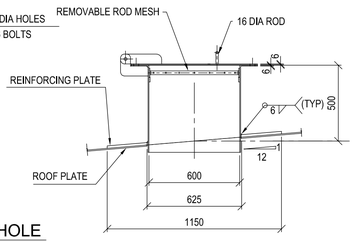
TYP. DETAIL OF SHELL DAVIT MANHOLE
SCALE- 1:20



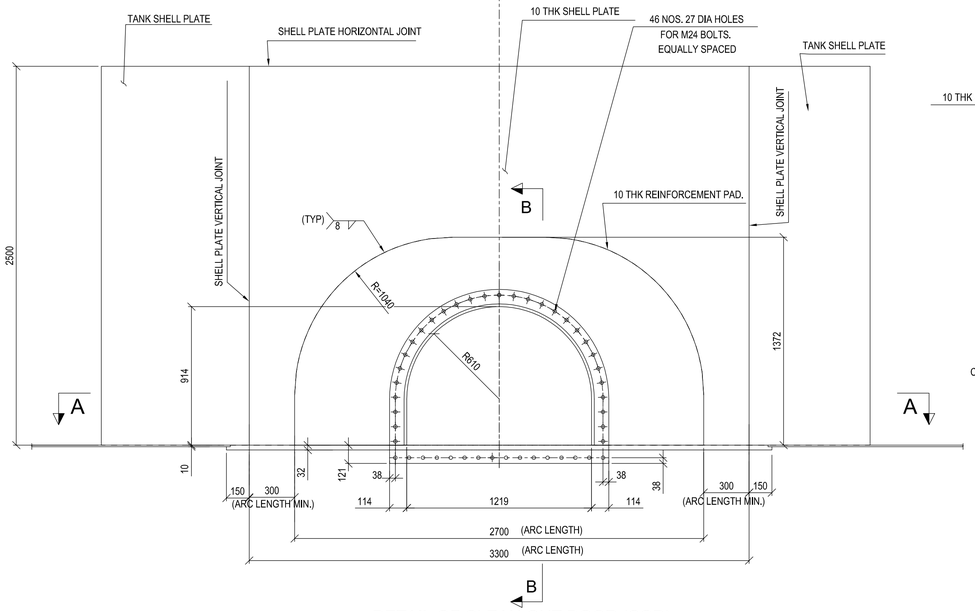
SECTION E-E
SCALE- 1:10



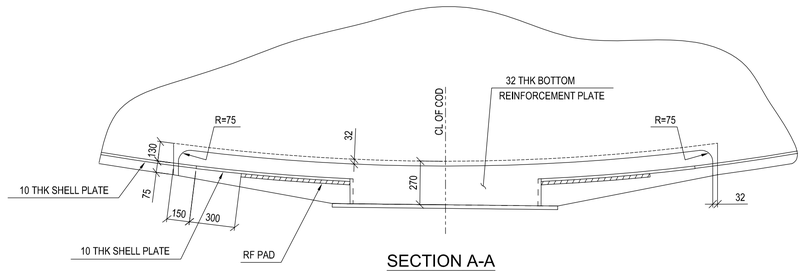
TYP. DETAIL OF ROOF MANHOLE
SCALE- 1:20



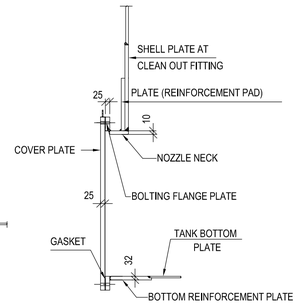
SECTION D-D
SCALE- 1:20



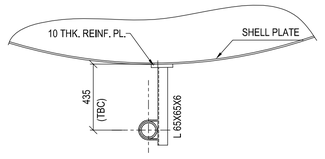
DETAIL OF CLEAN OUT DOOR (COD)
(SIZE 900x1200)
SCALE- 1:20



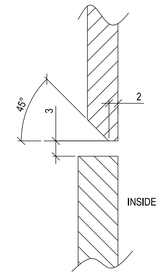
SECTION A-A
SCALE- 1:20



SECTION B-B
SCALE- 1:20



PLAN FOR INLET & OVERFLOW SUPPORT. FOR N1 & N12
SCALE- 1:20



TYP.SHELL PLATE JOINT DETAIL

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
3. FOR ALL OTHER NOTES REFER DRAWING MD - 216 - 3450 -EG -ST - GAD - 2245.
4. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
5. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3450 -EG -ST - GAD -2245 & 2247.
6. FOR LEGEND LIST REFER DRAWING MD - 216 - 3450 -EG -ST - GAD - 2245.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS.	D01	20.11.2023	AS BUILT	SJH	PRI	KDU	AG	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	C01	23/03/2020	ISSUED FOR CONSTRUCTION	SJH	PRI	KDU	AG	
MD-216-3450--EG-ME-GAD-1001	POX DEMIN WATER TANK 3450-TNK-001	A01	20/02/2020	ISSUED FOR REFERENCE	SJH	PRI	KDU	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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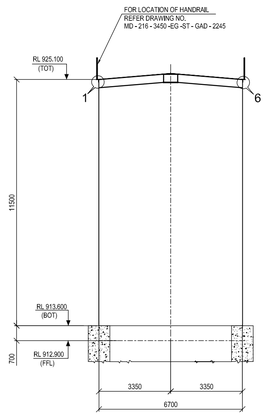
LARSEN & TOUBRO DRAWING NUMBER: 019045_450_S_G_26047_002 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707__GAU01__HH12__MTL01__34503 — DOCUMENT ID: OU100673485 — REVISION: 2

DRAWN	SJH	17/02/2020
CHECKED	PRI	19/02/2020
ENGINEER	KDU	18/02/2020
APPROVED	AG	20/02/2020
EMPLOYER		
SCALE	1:100,20	A1

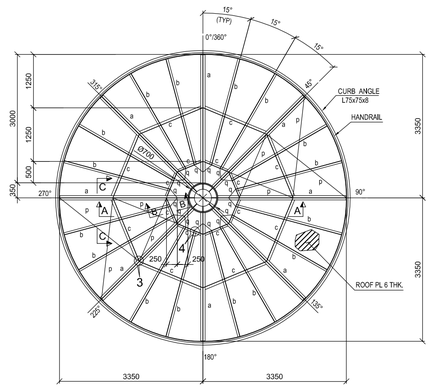
MANSOURAH-MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRESSURE OXIDATION-SERVICES
SHELL PLT DETAIL OF TANK 3450-TNK-001

DRAWING NUMBER: MD - 216 - 3450 - EG - ST - GAD - 2246 — REV: D01

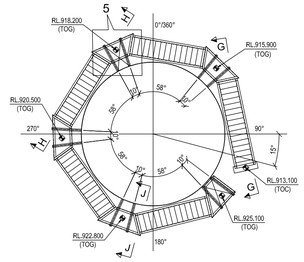
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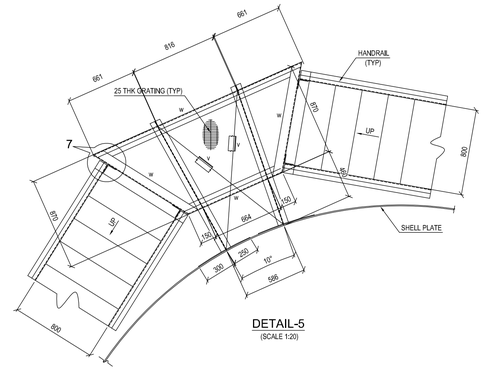
ELEVATION OF RAFTER
(SCALE 1:100)



PLAN AT ROOF LEVEL
(SCALE 1:50)



STAIR CASE PLAN
(SCALE 1:100)

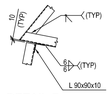


DETAIL-5
(SCALE 1:20)

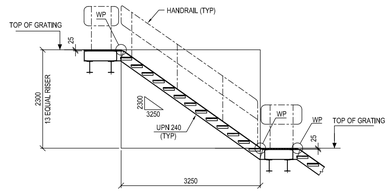
SCHEDULE OF MEMBERS

MEMBER MARKED	MEMBER	SECTION	FACTORED AXIAL FORCE(kN)	FACTORED SHEAR FORCE(kN)	REMARKS
a	IPE200	I	50	50	MATERIAL GRADE S275 JR
b	UPN120	[	20	-	
c	UPN 140	[	50	-	
p	L 65X65X6	⌐	30	-	
q	L 50X50X6	⌐	15	-	
t	IPE200	I	-	-	
u	L 65X65X6	⌐	-	-	
v	L 65X65X6	⌐	-	-	
w	UPN240	[	-	-	

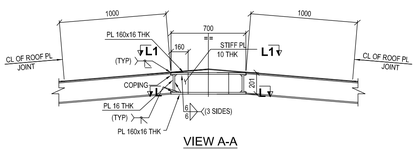
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED. MINIMUM 6 MM FILLET WELD OF 150 MM LENGTH SHALL BE PROVIDED. ALL GUSSET PLATE ARE 8 MM THK (UNO).



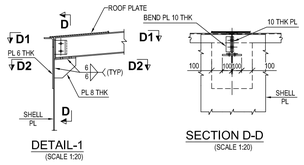
DETAIL-7
(SCALE 1:20)



SECTION H-H
ELEVATION OF STAIR
(SCALE 1:50)

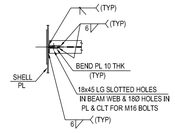


VIEW A-A
(TYP. CONNECTION OF MAIN RAFTER WITH CENTRAL DRUM)
(SCALE 1:20)

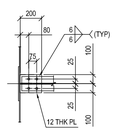


DETAIL-1
(SCALE 1:20)

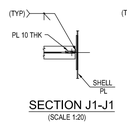
SECTION D-D
(SCALE 1:20)



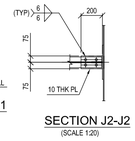
SECTION D1-D1
(SCALE 1:20)



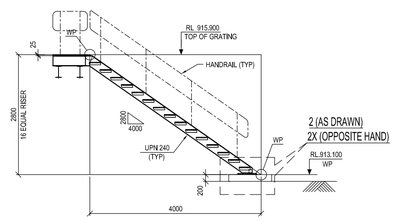
SECTION D2-D2
(SCALE 1:20)



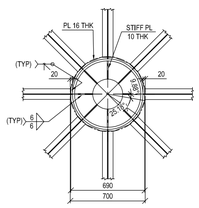
SECTION J1-J1
(SCALE 1:20)



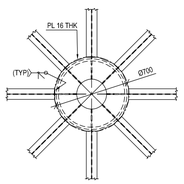
SECTION J2-J2
(SCALE 1:20)



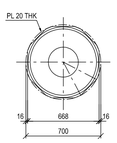
SECTION G-G
ELEVATION OF STAIR
(SCALE 1:50)



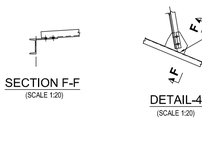
SECTION L-L
(SCALE 1:20)



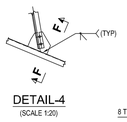
SECTION L1-L1
(SCALE 1:20)



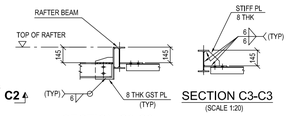
TOP & BOTTOM PLATE
(SCALE 1:20)



DETAIL-3
(SCALE 1:20)



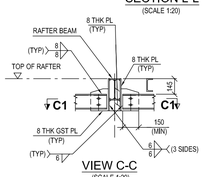
DETAIL-4
(SCALE 1:20)



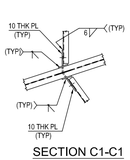
SECTION C2-C2
(SCALE 1:20)

SECTION C3-C3
(SCALE 1:20)

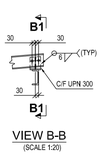
SECTION F-F
(SCALE 1:20)



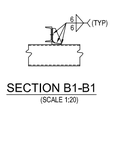
VIEW C-C
(SCALE 1:20)



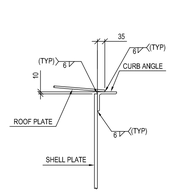
SECTION C1-C1
(SCALE 1:20)



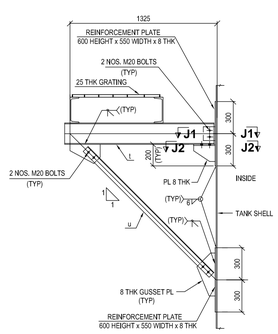
VIEW B-B
(SCALE 1:20)



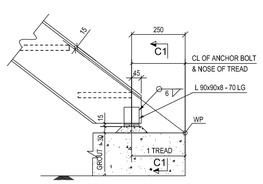
SECTION B1-B1
(SCALE 1:20)



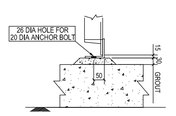
DETAIL-6
(TYP CONNECTION OF SHELL PL ROOF PL & CURB ANGLE)
(SCALE 1:10)



SECTION J-J
(SCALE 1:20)



DETAIL - 2 (AS DRAWN)
DETAIL - 2X (OPPOSITE HAND)
(SCALE 1:10)



SECTION C1-C1
(SCALE 1:10)

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
3. FOR ALL OTHER NOTES REFER DRAWING MD - 216 - 3450 -EG -ST - GAD - 2245.
4. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
5. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3450 -EG -ST - GAD - 2245 & 2246.
6. FOR LEGEND LIST REFER DRAWING MD - 216 - 3450 -EG -ST - GAD - 2245.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS.	D01	20.11.2023	AS BUILT	SJH	PRI	KDU	AG	
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL - DESIGN CRITERIA	C01	23/03/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	SJH	PRI	KDU	AG	
MD-216-3450-EG-ME-GAD-1001	P0X DEMIN WATER TANK 3450-TNK-001	A01	20/02/2020	ISSUED FOR REFERENCE	SJH	PRI	KDU	AG	

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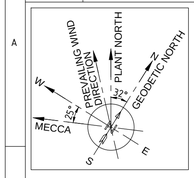
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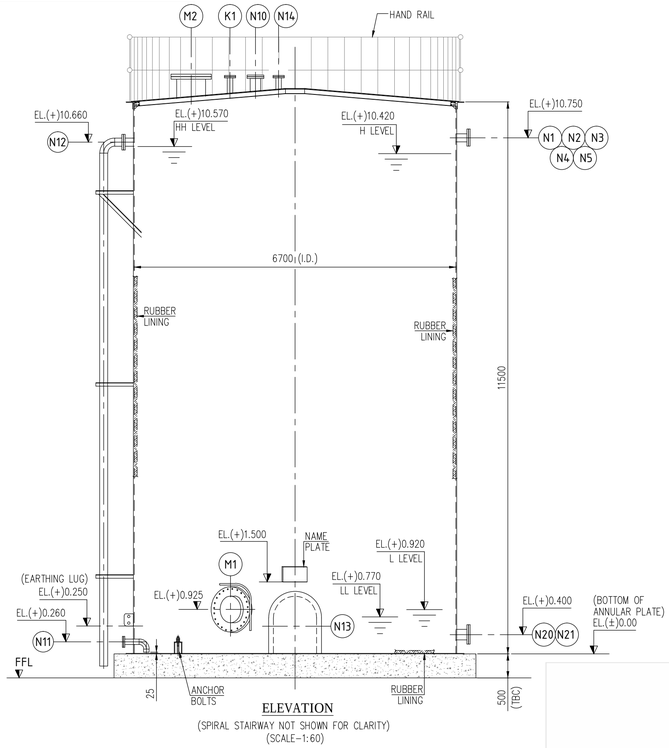
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LARSEN & TOUBRO DRAWING NUMBER: 01905_450_S_G_26047_003 — REVISION: 2
OUTOTEC DRAWING NUMBER: 900707__GAU01__HH12__MTL01__34504 — DOCUMENT ID: OU100673486 — REVISION: 2

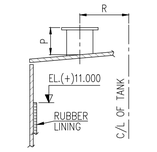
DRAWN	SJH	17/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	PRI	19/02/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	KDU	18/02/2020	PRESSURE OXIDATION-SERVICES
APPROVED	AG	20/02/2020	ROOF SYSTEM OF TANK 3450-TNK-001
EMPLOYER			
SCALE	1:200,100,20	A0	DRAWING NUMBER: MD - 216 - 3450 - EG - ST - GAD - 2247 — REV: D01

DRAWING NUMBER: MD-216-3450-EG-ME-GAD-1001 | REV: A01

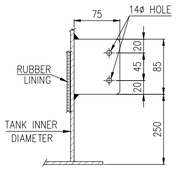




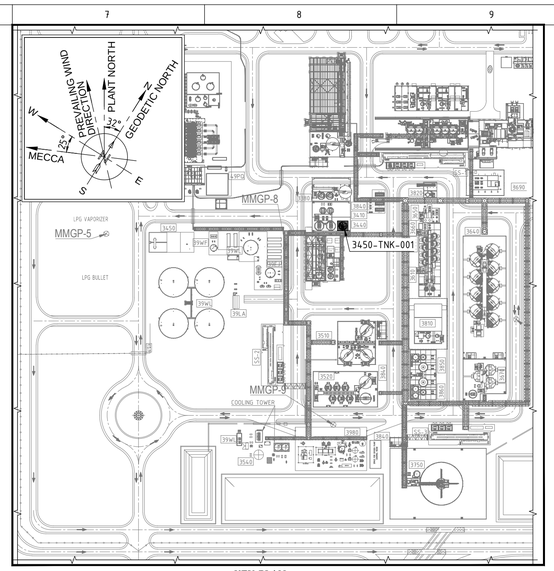
ELEVATION
(SPIRAL STAIRWAY NOT SHOWN FOR CLARITY)
(SCALE-1:60)



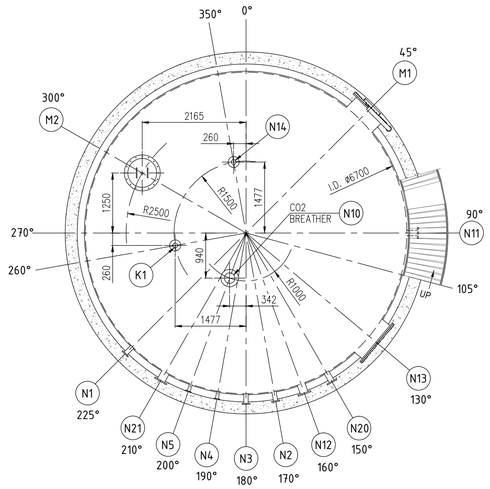
ROOF NOZZLES / FITTINGS



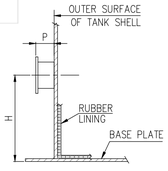
DETAIL OF EARTHING LUGS
(QUANTITY : 2 Nos.)



KEY PLAN
SCALE 1:2500



PLAN
3450-TNK-001
(SCALE-1:60)



SHELL NOZZLES / FITTINGS

NOTE:-
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METERS U.O.S.
2. ALL ELEVATION MEASUREMENTS ARE MEASURED FROM BOTTOM SIDE OF BASE PLATE.
3. SHELL NOZZLE ELEVATIONS ARE FROM BOTTOM OF BASE PLATE TO CENTERLINE OF NOZZLE.
4. SHELL NOZZLE PROJECTIONS ARE MEASURED FROM OUTSIDE OF SHELL TO FLANGE FACE.
5. ROOF NOZZLE PROJECTIONS ARE MEASURED FROM TOP OF ROOF SURFACE TO FLANGE FACE.
6. ROOF NOZZLE RADIUS MEASUREMENT ARE DISTANCES FROM TANK CENTERLINE TO NOZZLE CENTERLINE.
7. ALL NOZZLE LOCATION, ELEVATION AND ORIENTATION ARE INDICATIVE ONLY AND SHALL BE FINALIZED DURING DETAIL DESIGN PHASE OF PIPELINE ENGINEERING.
8. NOZZLE LOAD SHALL BE CONSIDERED & INCORPORATED AFTER COMPLETION OF PIPING ENGINEERING.
9. LOCATION OF EARTHING LUGS SHALL BE SHOWN IN SEPARATE DRAWING.
10. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY.
11. ANCHOR BOLTS SHOWN HERE IS INDICATIVE, QUANTITY & LOCATION SHALL BE SHOWN IN SEPARATE DRAWING.

ABBREVIATIONS / LEGENDS

ID	- INNER DIAMETER
COD	- CLEAN OUT DOOR
EL	- ELEVATION
RF	- REINFORCEMENT
TBC	- TO BE CONFIRMED
U.O.S	- UNLESS OTHERWISE STATED
H	- HIGH
HH	- HIGH HIGH
L	- LOW
LL	- LOW LOW
RFSO	- RAISED FACE SLIP ON
C/W	- COUNTER WITH
FFL	- FINISH FLOOR LEVEL
PWHT	- POST WELD HEAT TREATMENT

MATERIAL SPECIFICATION

SHELL, BOTTOM, ANNULAR, ROOF	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
COD	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
MANWAY NECKS/ MANWAY FLANGES	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
NOZZLE NECKS (PIPES) (600 NB & BELOW)	ASTM A53 Gr.B/ ASTM A106 Gr.B
NOZZLE NECKS (PIPES) (> 600 NB)	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
NOZZLE FLANGES/ BLIND FLANGE	SA 105
GASKETS FOR NOZZLES	RING TYPE (FLAT RING) 1.5mm THICK COMPRESSED ASBESTOS FIBRE. Prohibited material
GASKETS FOR MANWAY/ COD	3mm THICK COMPRESSED NON-ASBESTOS SHEETS.
BOLTS/ NUTS FOR NOZZLE	SA 193 Gr. B7 / 194 Gr. 2H
BOLTS/ NUTS FOR MANWAY	SA 193 Gr. B7 / 194 Gr. 2H
STRUCTURAL BOLTS	ASTM A325M
ANCHOR CHAIR	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
STRUCTURAL(ROLLED SECTION)-STAIRWAY / PLATFORM / ROOF SUPPORTING STRUCTURE	ASTM A36 GR.36 OR EN 10025 GR.S275 JR/S355 JR
HANDRAIL PIPE / HANDRAIL POST	ASTM A53 Gr. B
PIPE FITTINGS	SA 234 Gr. WPB
REINFORCEMENT PAD/GUSSET/ EXTERNAL SUPPORT ETC.	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
EARTHING LUGS	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
ANCHOR BOLTS	ASTM F1554
CURB ANGLE	ASTM A36 GR.36 OR EN 10025 GR.S275 JR/S355 JR
NAME PLATE	STAINLESS STEEL (SS 316)

DESIGN DATA

LOCATION		MANSOURAH - MASSARAH GOLD PROJECT
EQUIPMENT TAG NO.		3450-TNK-001
TANK NAME		POX DEMIN WATER TANK
FIELD ERECTED/SHOP FABRICATED		FIELD ERECTED
DESIGN CODE & EDITION		API 650, 12TH EDITION, ADDENDUM 3, AUGUST 2018
DESIGN DATA - PROCESS		
STORED FLUID		DEMINERALIZED WATER
DENSITY OF CONTENT (t/m³)	NORMAL OPERATION / DESIGN	1.0 / 1.0
PRESSURE (kPa)	VACUUM / DESIGN	ATM / ATM
TEMPERATURE (°C)	NORMAL OPERATION / DESIGN	22 / 50
DENSITY OF SCALE (t/m³)		N/A
DENSITY SCALE THICKNESS (mm)		N/A
OPERATING LEVELS (mm)	MINIMUM / MAXIMUM	760 / 10560
DESIGN FILL RATE (m³/h)		100 (TBC)
DESIGN EMPTY RATE (m³/h)		80 (TBC)
PWHT REQUIRED (YES/NO). IF YES SPECIFY REASON FOR PWHT		NO
DESIGN DATA - MECHANICAL		
TANK DIAMETER (I.D.) (mm)		6700
TANK CYLINDRICAL HEIGHT (mm)		11500
NET WORKING CAPACITY (m³)		335
MAXIMUM CAPACITY (m³)		372
TANK HAS SLOPED BOTTOM	(YES/NO)	YES
	BOTTOM SLOPE	1/120 CONICAL APEX-UP
ROOF REQUIRED	ROOF TYPE	SUPPORTED CONE ROOF
	ROOF SLOPE	1 IN 12
	ROOF SUPPORT	SUPPORTED ON RAFTER/TRUSS
	ROOF UNIFORM LIVE LOAD(kPa)	1.2
AGITATOR / RAKE REQUIRED	(YES/NO)	NO
	EQUIPMENT TYPE(AGITATOR/RAKE)	N/A
	AGITATOR/RAKE SUPPORTED BY TANK (YES/NO)	N/A
CORROSION ALLOWANCE (mm)	SHELL / ROOF / BOTTOM	1.6 / 0 /1.6
	STRUCTURAL	1.6

DESIGN DATA - OTHERS

SEISMIC USE GROUP, SUG PER API650 E.3.1		SUG I
FOUNDATION TYPE		RING WALL
INSULATION REQUIRED	(YES/NO)	NO
	MATERIAL	N/A
	THICKNESS(mm) ROOF/SHELL/BOTTOM	N/A
LINER REQUIRED	(YES/NO)	YES (UPTO 11.0 M HEIGHT)
	LINER MATERIAL	NATURAL RUBBER SHORE HARDNESS 65±5A
	LINER THICKNESS (mm) SHELL/BOTTOM	6.0 mm.
PAINTING	EXTERNAL	YES
	INTERNAL	YES (UNLINED AREA)

TANK APPURTENANCE (AS REQUIRED)

PIPE SUPPORT	YES
SPIRAL STAIRWAY	YES
ROOF CIRCUMFERENTIAL HANDRAIL	YES
WIND GIRDER	AS REQUIRED
CATHODIC PROTECTION	NO
LEAK DETECTION	NO
INTERNAL BAFFLES	NO
EARTHING LUGS	YES

NOZZLE TABLE

NOZZLE	SERVICE	SIZE (DN)	THICKNESS (SCHEDULE)	RATING #	TYPE	ELEVATION "H"	PROJECTION "P"	RADIUS "R"	ORIENTATION	LOCATION	REMARK
N1	INLET	150	Sch XS	CL 150	RFSO	10.750	200	-	225°	SHELL	
N2	RE-CIRCULATION LINE FOR POX DEMIN WATER PUMP	80	Sch XS	CL 150	RFSO	10.750	175	-	170°	SHELL	
N3	COMBINED RE-CIRCULATION LINE FOR HP DEMIN WATER PUMP	80	Sch XS	CL 150	RFSO	10.750	175	-	180°	SHELL	
N4	RE-CIRCULATION LINE FOR HP DEMIN WATER PUMP	50	Sch XS	CL 150	RFSO	10.750	150	-	190°	SHELL	
N5	RE-CIRCULATION LINE FOR HP DEMIN WATER PUMP	50	Sch XS	CL 150	RFSO	10.750	150	-	200°	SHELL	
N10	VENT/CO2 BREATHER	150(TBC)	-	CL 150	RFSO	-	300	1000	200°	ROOF	
N11	DRAIN	100	Sch XS	CL 150	RFSO	0.260	200		90°	SHELL	
N12	OVERFLOW	150	Sch XS	CL 150	RFSO	10.660	200	-	160°	SHELL	
N13	CLEAN OUT DOOR (COD)-hxb	900x1200	-	AP1 650		-	-	-	130°	SHELL	
N14	SPARE C/W BLIND	100	Sch XS	CL 150	RFSO	-	400	1500	350°	ROOF	
N20	OUTLET-POX DEMIN WATER	100	Sch XS	CL 150	RFSO	0.350	200	-	150°	SHELL	
N21	OUTLET-HP DEMIN WATER	150	Sch XS	CL 150	RFSO	0.350	200	-	210°	SHELL	
K1	LEVEL TRANSMITTER	100	Sch XS	CL 150	RFSO	-	400	1500	260°	ROOF	
M1	MANHOLE C/W DAVIT	750	-	AP1 650		0.925	175	-	45°	SHELL	
M2	MANHOLE C/W HINGED COVER	600	-			-	500	2500	300°	ROOF	

Document Number MD-216-3450-EG-ME-GAD-1001
Revision A01
Approved with Comments
Mr Doron Golan - Bechtel Global
Feb 20, 2020, 1:11 PM GMT+2:00
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DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3450-EG-ME-SPC-1001	SPECIFICATION : POX DEMIN WATER TANK								
MD-216-3450-EG-PR-PID-1020	PIPING INSTRUMENTATION DIAGRAM (PID)								
MD-216-3450-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM (PFD)	A01	21/01/2020	ISSUED FOR APPROVAL	SUG	SPU	SPU	KBJ	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO DRAWING NUMBER: O19045_080_U_G_20017_001 | REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _HH12 _MTL01 _34501 | DOCUMENT ID: OU100667074 | REVISION: 0

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DRAWN	SUG	02/01/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SPU	10/01/2020	GENERAL ARRANGEMENT
ENGINEER	SPU	10/01/2020	PRESSURE OXIDATION-SERVICES
APPROVED	KBJ	21/01/2020	POX DEMIN WATER TANK 3450-TNK-001
EMPLOYER			
SCALE	1:60	A1	DRAWING NUMBER: MD - 216 - 3450 - EG - ME - GAD - 1001 | REV: A01

ATTACHMENT 3 STAAD OUTPUT

PROPRIETARY & CONFIDENTIAL

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*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    JUN 10, 2020                 *
*          Time=    12:28:27                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
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INPUT FILE: TANK FDN (3450).STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 10.06.2020
    4. JOB NAME 3450-TNK-001
    5. JOB CLIENT MA'ADEN
    6. ENGINEER NAME SOR
    7. CHECKER NAME SJM
    8. APPROVED NAME AG
    9. JOB REV 0
   10. END JOB INFORMATION
   11. INPUT WIDTH 79
   12. UNIT MMS KN
   13. JOINT COORDINATES
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152. 7742 3535 3553 3359 3360; 7743 3553 3529 3358 3359
153. ELEMENT PROPERTY
154. 7386 TO 7481 THICKNESS 600
155. 7600 TO 7667 7720 TO 7743 THICKNESS 500
156. DEFINE MATERIAL START
157. ISOTROPIC CONCRETE
158. E 21.7185
159. POISSON 0.17
160. DENSITY 2.35616E-008
161. ALPHA 1E-005
162. DAMP 0.05
163. TYPE CONCRETE
164. STRENGTH FCU 0.027579
165. END DEFINE MATERIAL
166. CONSTANTS
167. MATERIAL CONCRETE ALL
168. START GROUP DEFINITION
169. ELEMENT
170. _WALL 7386 TO 7481
171. _RAFT 7600 TO 7667 7720 TO 7743
172. END GROUP DEFINITION
173. UNIT METER KN
174. SUPPORTS
175. 7600 TO 7667 7720 TO 7743 PLATE MAT DIRECT Y SUBGRADE 36000
176. ********************************************************************************
177. LOAD 1 LOADTYPE DEAD  TITLE DEAD LOAD ( D )
178. SELFWEIGHT Y -1
179. ELEMENT LOAD
180. 7600 TO 7667 PR GY -29
181. 7720 TO 7743 PR GY -17
182. JOINT LOAD
183. 3452 TO 3475 FY -13.42
184. ELEMENT LOAD
185. 7386 TRAP JT 10.875 10.875 14.5 14.5
186. 7387 TRAP JT 10.875 10.875 14.5 14.5
187. 7388 TRAP JT 10.875 10.875 14.5 14.5
188. 7389 TRAP JT 10.875 10.875 14.5 14.5
189. 7390 TRAP JT 10.875 10.875 14.5 14.5
190. 7391 TRAP JT 10.875 10.875 14.5 14.5
191. 7392 TRAP JT 10.875 10.875 14.5 14.5
192. 7393 TRAP JT 10.875 10.875 14.5 14.5
193. 7394 TRAP JT 10.875 10.875 14.5 14.5
194. 7395 TRAP JT 10.875 10.875 14.5 14.5
195. 7396 TRAP JT 10.875 10.875 14.5 14.5
196. 7397 TRAP JT 10.875 10.875 14.5 14.5
197. 7398 TRAP JT 10.875 10.875 14.5 14.5
198. 7399 TRAP JT 10.875 10.875 14.5 14.5
199. 7400 TRAP JT 10.875 10.875 14.5 14.5
200. 7401 TRAP JT 10.875 10.875 14.5 14.5
201. 7402 TRAP JT 10.875 10.875 14.5 14.5
202. 7403 TRAP JT 10.875 10.875 14.5 14.5
203. 7404 TRAP JT 10.875 10.875 14.5 14.5
204. 7405 TRAP JT 10.875 10.875 14.5 14.5
205. 7406 TRAP JT 10.875 10.875 14.5 14.5
206. 7407 TRAP JT 10.875 10.875 14.5 14.5
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207. 7408 TRAP JT 10.875 10.875 14.5 14.5
208. 7409 TRAP JT 10.875 10.875 14.5 14.5
209. 7410 TRAP JT 7.25 7.25 10.875 10.875
210. 7411 TRAP JT 7.25 7.25 10.875 10.875
211. 7412 TRAP JT 7.25 7.25 10.875 10.875
212. 7413 TRAP JT 7.25 7.25 10.875 10.875
213. 7414 TRAP JT 7.25 7.25 10.875 10.875
214. 7415 TRAP JT 7.25 7.25 10.875 10.875
215. 7416 TRAP JT 7.25 7.25 10.875 10.875
216. 7417 TRAP JT 7.25 7.25 10.875 10.875
217. 7418 TRAP JT 7.25 7.25 10.875 10.875
218. 7419 TRAP JT 7.25 7.25 10.875 10.875
219. 7420 TRAP JT 7.25 7.25 10.875 10.875
220. 7421 TRAP JT 7.25 7.25 10.875 10.875
221. 7422 TRAP JT 7.25 7.25 10.875 10.875
222. 7423 TRAP JT 7.25 7.25 10.875 10.875
223. 7424 TRAP JT 7.25 7.25 10.875 10.875
224. 7425 TRAP JT 7.25 7.25 10.875 10.875
225. 7426 TRAP JT 7.25 7.25 10.875 10.875
226. 7427 TRAP JT 7.25 7.25 10.875 10.875
227. 7428 TRAP JT 7.25 7.25 10.875 10.875
228. 7429 TRAP JT 7.25 7.25 10.875 10.875
229. 7430 TRAP JT 7.25 7.25 10.875 10.875
230. 7431 TRAP JT 7.25 7.25 10.875 10.875
231. 7432 TRAP JT 7.25 7.25 10.875 10.875
232. 7433 TRAP JT 7.25 7.25 10.875 10.875
233. 7434 TRAP JT 3.625 3.625 7.25 7.25
234. 7435 TRAP JT 3.625 3.625 7.25 7.25
235. 7436 TRAP JT 3.625 3.625 7.25 7.25
236. 7437 TRAP JT 3.625 3.625 7.25 7.25
237. 7438 TRAP JT 3.625 3.625 7.25 7.25
238. 7439 TRAP JT 3.625 3.625 7.25 7.25
239. 7440 TRAP JT 3.625 3.625 7.25 7.25
240. 7441 TRAP JT 3.625 3.625 7.25 7.25
241. 7442 TRAP JT 3.625 3.625 7.25 7.25
242. 7443 TRAP JT 3.625 3.625 7.25 7.25
243. 7444 TRAP JT 3.625 3.625 7.25 7.25
244. 7445 TRAP JT 3.625 3.625 7.25 7.25
245. 7446 TRAP JT 3.625 3.625 7.25 7.25
246. 7447 TRAP JT 3.625 3.625 7.25 7.25
247. 7448 TRAP JT 3.625 3.625 7.25 7.25
248. 7449 TRAP JT 3.625 3.625 7.25 7.25
249. 7450 TRAP JT 3.625 3.625 7.25 7.25
250. 7451 TRAP JT 3.625 3.625 7.25 7.25
251. 7452 TRAP JT 3.625 3.625 7.25 7.25
252. 7453 TRAP JT 3.625 3.625 7.25 7.25
253. 7454 TRAP JT 3.625 3.625 7.25 7.25
254. 7455 TRAP JT 3.625 3.625 7.25 7.25
255. 7456 TRAP JT 3.625 3.625 7.25 7.25
256. 7457 TRAP JT 3.625 3.625 7.25 7.25
257. 7458 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
258. 7459 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
259. 7460 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
260. 7461 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
261. 7462 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
262. 7463 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
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263. 7464 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
264. 7465 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
265. 7466 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
266. 7467 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
267. 7468 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
268. 7469 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
269. 7470 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
270. 7471 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
271. 7472 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
272. 7473 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
273. 7474 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
274. 7475 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
275. 7476 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
276. 7477 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
277. 7478 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
278. 7479 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
279. 7480 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
280. 7481 TRAP JT 9.98974E-005 9.98974E-005 3.625 3.625
281. ******************************************************************************
282. LOAD 2 LOADTYPE FLUIDS  TITLE OPERATING LIQUID LOAD ( F1 )
283. ELEMENT LOAD
284. 7600 TO 7667 PR GY -103.75
285. 7386 TO 7481 PR 51.87
286. ******************************************************************************
287. LOAD 3 LOADTYPE FLUIDS  TITLE TEST LOAD ( F2 )
288. ELEMENT LOAD
289. 7600 TO 7667 PR GY -112.86
290. 7386 TO 7481 PR 56.43
291. ******************************************************************************
292. LOAD 4 LOADTYPE WIND  TITLE WIND LOAD ( W )
293. JOINT LOAD
294. 3452 TO 3475 FZ -1.75
295. 3452 FY -9.2
296. 3453 FY -8.8
297. 3454 FY -7.9
298. 3455 FY -6.5
299. 3456 FY -4.6
300. 3457 FY -2.4
301. 3458 FY 0
302. 3459 FY 2.4
303. 3460 FY 4.6
304. 3461 FY 6.5
305. 3462 FY 7.9
306. 3463 FY 8.8
307. 3464 FY 9.2
308. 3465 FY 8.8
309. 3466 FY 7.9
310. 3467 FY 6.5
311. 3468 FY 4.6
312. 3469 FY 2.4
313. 3470 FY 0
314. 3471 FY -2.4
315. 3472 FY -4.6
316. 3473 FY -6.5
317. 3474 FY -7.9
318. 3475 FY -8.8
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319. ********************************************************************************
320. LOAD 5 LOADTYPE SEISMIC  TITLE SEISMIC LOAD ( E )
321. JOINT LOAD
322. 3452 TO 3475 FZ -2.542
323. 3452 FY -7.2
324. 3453 FY -7
325. 3454 FY -6.3
326. 3455 FY -5.1
327. 3456 FY -3.6
328. 3457 FY -1.9
329. 3458 FY 0
330. 3459 FY 1.9
331. 3460 FY 3.6
332. 3461 FY 5.1
333. 3462 FY 6.3
334. 3463 FY 7
335. 3464 FY 7.2
336. 3465 FY 7
337. 3466 FY 6.3
338. 3467 FY 5.1
339. 3468 FY 3.6
340. 3469 FY 1.9
341. 3470 FY 0
342. 3471 FY -1.9
343. 3472 FY -3.6
344. 3473 FY -5.1
345. 3474 FY -6.3
346. 3475 FY -7
347. ********************************************************************************
348. LOAD 6 LOADTYPE FLUIDS  TITLE LIVE LOAD ( L )
349. JOINT LOAD
350. 3452 TO 3475 FY -6.21
351. ********************************************************************************
352. * LOAD COMBINATIONS
353. LOAD COMB 201 1.4 ( D + F1 )
354. 1 1.4 2 1.4
355. LOAD COMB 202 1.4 ( D + F1 ) + 1.7 L
356. 1 1.4 2 1.4 6 1.7
357. LOAD COMB 203 0.9 D + 1.6 W
358. 1 0.9 4 1.6
359. LOAD COMB 204 1.2 ( D + F1 ) + 1.6 W + L
360. 1 1.2 2 1.2 4 1.6 6 1.0
361. LOAD COMB 205 1.2 ( D + F1 ) + E + L
362. 1 1.2 2 1.2 5 1.0 6 1.0
363. LOAD COMB 206 0.9 ( D + F1 ) + E
364. 1 0.9 2 0.9 5 1.0
365. LOAD COMB 207 1.4 ( D + F2 )
366. 1 1.4 3 1.4
367. LOAD COMB 208 1.4 ( D + F2 ) + 1.7 L
368. 1 1.4 3 1.4 6 1.7
369. LOAD COMB 209 1.2 ( D + F2 ) + 1.6 W + L
370. 1 1.2 3 1.2 4 1.6 6 1.0
371. LOAD COMB 210 1.2 ( D + F2 ) + E + L
372. 1 1.2 3 1.2 5 1.0 6 1.0
373. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 195 NUMBER OF MEMBERS 0
NUMBER OF PLATES 188 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 99

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 192/ 27/ 132 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 873
TOTAL LOAD COMBINATION CASES = 10 SO FAR.
SIZE OF STIFFNESS MATRIX = 116 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 14.3/ 145562.8 MB

374. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= JUN 10,2020 TIME= 12:28:33 ****

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*   http://selectservices.bentley.com/en-US/               *
*                                                          *
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*   Bentley and Partners can be found at program menu      *
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